

CDKN2B fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes CDKN2B gene fusions from the msk_impact_50k_2026 study, covering 12 fusion events. 2/12 fusions (16.7%) were in-frame. 2/12 fusions (16.7%) retained the PF12796 domain. Domain retention was tested with Fisher's exact test ($p=0.0222222$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for CDKN2B.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 12 analyzed fusion events, identifying 7 distinct fusion partner genes. The most frequent partner was CDKN2A, observed in 6 events.

1.2 Domain retention

PF12796 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/2 (100.0%) of in-frame fusions retained the domain, $p=0.0222222$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

1.3 Domain disruption

Domain-disruption analysis was skipped: CDKN2B has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 10 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 52 aa (permutation-corrected $p=0.762376$, not statistically significant at $\alpha=0.05$). This is -2 aa from the nearest configured domain boundary at 54 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: CDKN2B has no `group_field` configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p-value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: CDKN2B has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CDKN2A	6	0.5	0.20043213737826426		0.6741935483870968		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.43033866657518655	1
CNTRL	1	0.08333333333333333	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.32158290620652297	2
MTAP	1	0.08333333333333333	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.32158290620652297	3
SLC24A2	1	0.08333333333333333	0.20043213737826426		0.7741935483870968		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2731958094323294	4
GAB2	1	0.08333333333333333	0.20043213737826426		0.6612903225806452		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24900226104523265	5
ELAVL2	1	0.08333333333333333	0.20043213737826426				['domain_retention', 'recurrence']	0.13656006244466556	6
CDKN2B-AS1	1	0.08333333333333333	0.20043213737826426		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.10729719192080867	7

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
38	0	2	1	7	0.0	1.0	-0.0
52	2	0	3	5		0.4444444444444453	0.3521825181113624
69	2	0	4	4		0.4666666666666668	0.3309932190414243
73	2	0	5	3		1.0	-0.0
75	2	0	6	2		1.0	-0.0
113	2	0	7	1		1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

2	0
0	8

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
auto_key_domain	Ankyrin repeats (3 copies)	10	2	5	3	0.34104938271604934

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
CDKN2A	6
CDKN2B-AS1	1
CNTRL	1
ELAVL2	1
GAB2	1
MTAP	1
SLC24A2	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	probabilistic_breakpoint	is_disrupted	domain_retention_fraction	domain_lost_fraction	domain_disrupted_fraction	disrupted_domain	source_annotation	effect_on_frame
EVT-P-0035576-T01-IM6-2	P-0035576-T01-IM6		CDKN2A-CDKN2B	CDKN2A	unknown	five_prime	22006179	2	75	False	disrupted	0.2716049382716049	0.2716049		Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_000077) Rearrangement: c.224:CDKN2B_c.153:CDKN2Adel Protein Fusion: mid-exon {CDKN2B:CDKN2A} CDKN2B NA (NM_004936) - CDKN2A (NM_058195) rearrangement: c.206:CDKN2B_c.194-11:CDKN2Adel Protein Fusion: out of frame {CDKN2B:CDKN2A}	CDKN2A
EVT-P-0010183-T02-IM6-10	P-0010183-T02-IM6		CDKN2B-CDKN2A	CDKN2A	unknown	five_prime	22006197	2	69	False	disrupted	0.19753086419753085	0.19753085		Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_058195) rearrangement: c.206:CDKN2B_c.194-11:CDKN2Adel Protein Fusion: mid-exon {CDKN2B:CDKN2A} CDKN2B NA (NM_004936) - CDKN2A (NM_000077) rearrangement: c.156+629:CDKN2B_c.458-143:CDKN2Adel Protein Fusion: out of frame {CDKN2B:CDKN2A}	CDKN2A
EVT-P-0060581-T01-IM7-11	P-0060581-T01-IM7		CDKN2B-CDKN2A	CDKN2A	out-of-frame	five_prime	22008168	1	52	True	lost	0.0	False		Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_000077) rearrangement: c.156+629:CDKN2B_c.458-143:CDKN2Adel Protein Fusion: out of frame {CDKN2B:CDKN2A}	CDKN2A

3 Figures

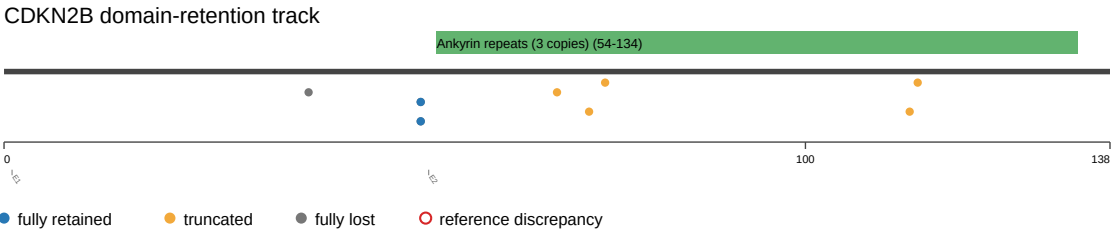


Figure 1: domain retention outliers

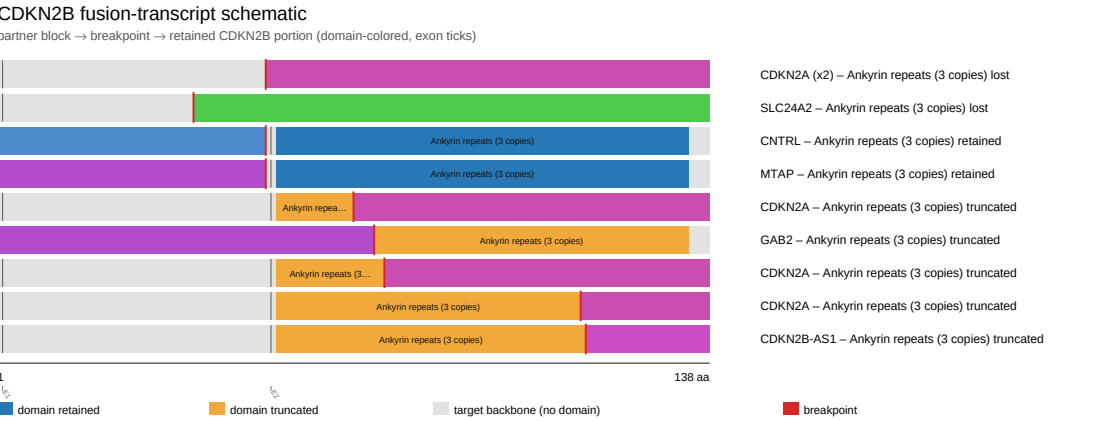


Figure 2: fusion schematic

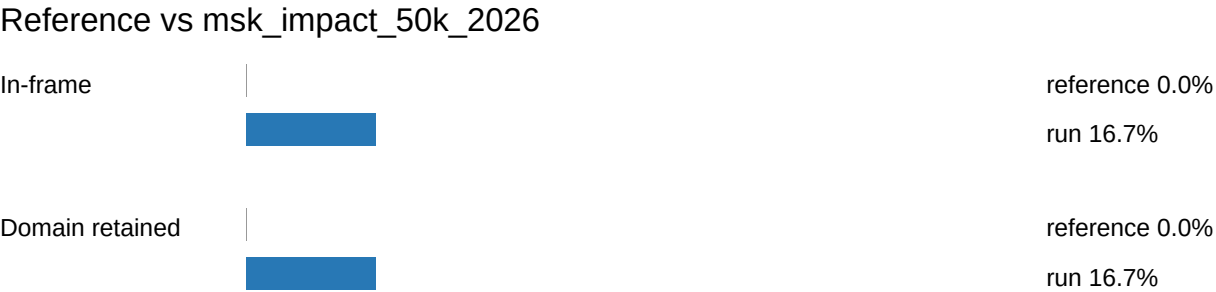


Figure 3: reference comparison